

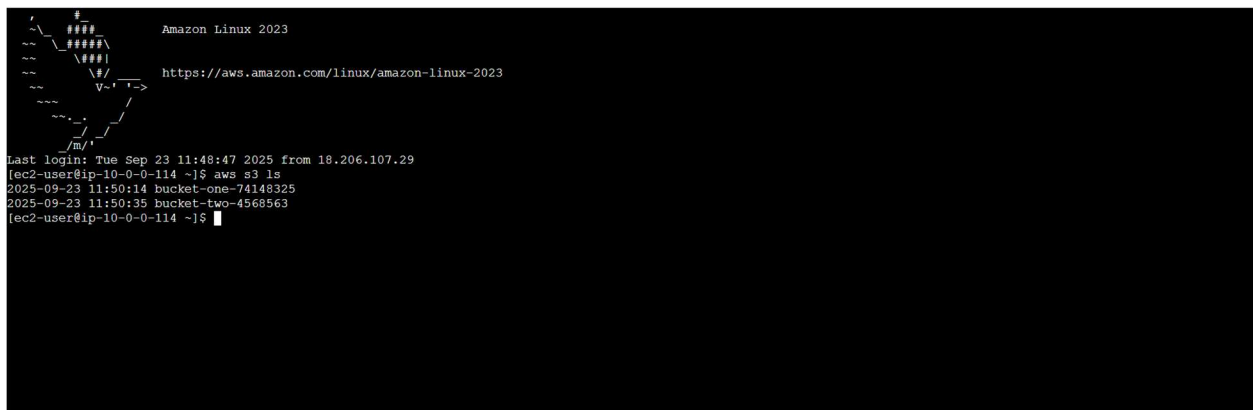
## 1) Internet Gateway (IGW) → For public internet access

### Steps:

1. Go to **VPC** → **Internet Gateways** → **Create IGW**.
2. **Attach IGW** to your VPC.
3. Open your **Route Table (public subnet)** → Add route `0.0.0.0/0` → **IGW**.
4. Ensure your **EC2 has a public IP** (auto-assign or Elastic IP).
5. attach an **IAM Role** with an S3 policy to the EC2 instance.

\* Now EC2 can access the internet

### Test S3 Access

A terminal window on an Amazon Linux 2023 EC2 instance. The prompt is [ec2-user@ip-10-0-0-114 ~]\$. The user runs 'aws s3 ls' and the output shows two buckets: 'bucket-one-74148325' and 'bucket-two-4568563'. The terminal also shows the last login time and the public/private IP addresses.

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Tue Sep 23 11:48:47 2025 from 18.206.107.29
[ec2-user@ip-10-0-0-114 ~]$ aws s3 ls
2025-09-23 11:50:14 bucket-one-74148325
2025-09-23 11:50:35 bucket-two-4568563
[ec2-user@ip-10-0-0-114 ~]$
```

i-024e56489b4c1eca8 (Public-ec2)

PublicIPs: 34.226.222.76 PrivateIPs: 10.0.0.114

Activate Windows

\* Now I am able to see bucket list

## 2) NAT Gateway → For private subnet EC2 to reach internet/AWS services

### Steps:

1. Create an **NAT Gateway**.
2. Go to **VPC** → **NAT Gateways** → **Create NAT Gateway** → Place it in a **public subnet**, attach the Elastic IP.
3. Open **Route Table (private subnet)** → Add route `0.0.0.0/0` → **NAT Gateway**.
4. Launch EC2 in the **private subnet (no public IP)**.
5. attach an **IAM Role** with an S3 policy to the EC2 instance.

\* Now private EC2 can reach the internet

## Test S3 Access

From Bastion, connect to Private EC2

```
#
~\  #####
~\  #####\
~\  #####!
~\  \##/
~\  V~' ' ->
~\  /m/'
~\  /m/'

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Tue Sep 23 11:50:59 2025 from 18.206.107.27
[ec2-user@ip-10-0-0-114 ~]$ ssh -i "test.pem" ec2-user@10.0.1.176
#
~\  #####
~\  #####\
~\  #####!
~\  \##/
~\  V~' ' ->
~\  /m/'
~\  /m/'

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Tue Sep 23 11:58:30 2025 from 10.0.0.114
[ec2-user@ip-10-0-1-176 ~]$
```

## Test S3 Access

```
#
~\  #####
~\  #####\
~\  #####!
~\  \##/
~\  V~' ' ->
~\  /m/'
~\  /m/'

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Tue Sep 23 12:02:29 2025 from 18.206.107.28
[ec2-user@ip-10-0-0-114 ~]$ ssh -i "test.pem" ec2-user@10.0.1.176
#
~\  #####
~\  #####\
~\  #####!
~\  \##/
~\  V~' ' ->
~\  /m/'
~\  /m/'

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Tue Sep 23 12:02:34 2025 from 10.0.0.114
[ec2-user@ip-10-0-1-176 ~]$ aws s3 ls
2025-09-23 11:50:14 bucket-one-74148325
2025-09-23 11:50:35 bucket-two-4568563
[ec2-user@ip-10-0-1-176 ~]$
```

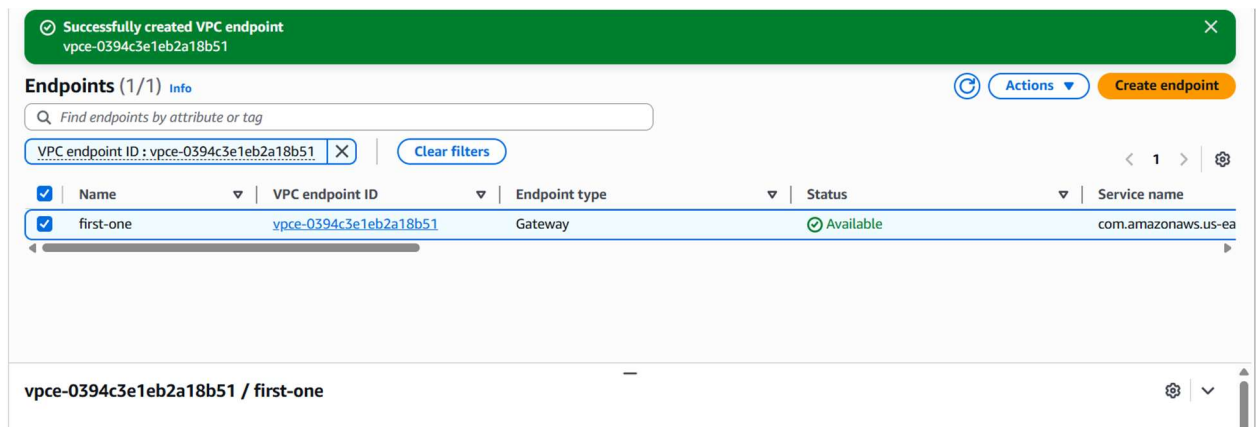
\* Now I am able to access s3 in private instance

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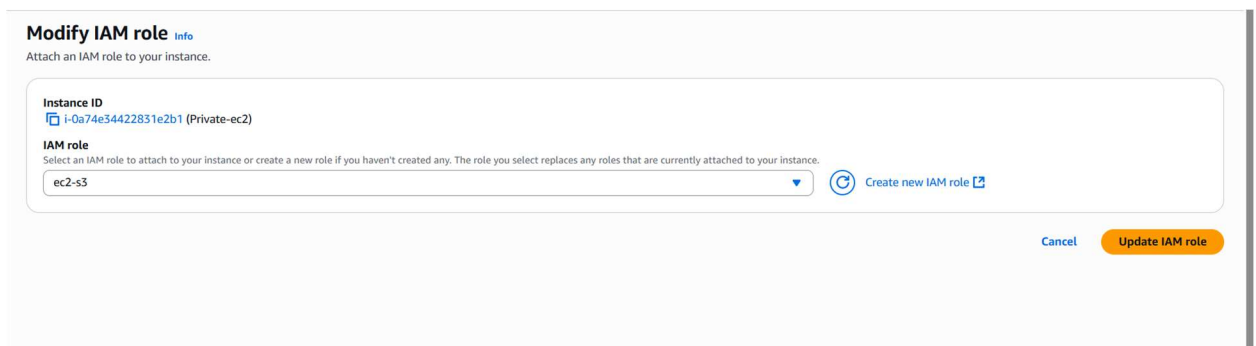
### 3) VPC Endpoints → For private access to AWS services (like S3)

#### Steps:

1. Go to **VPC → Endpoints → Create Endpoint**.
2. Choose service:
  - **Gateway Endpoint** for S3.
  - **Interface Endpoint** for most other AWS services (EC2 API, Secrets Manager, etc.).
3. Select your **VPC** and associate the correct **Route Tables** (for Gateway) or **Subnets + Security Groups** (for Interface).
4. Create the endpoint.



5. attach an **IAM Role** with an S3 policy to the EC2 instance.



\*Now EC2 can connect to AWS services without Internet Gateway or NAT Gateway.(delete your old nat gateway )

|                               |                                       |        |                                          |   |                             |           |
|-------------------------------|---------------------------------------|--------|------------------------------------------|---|-----------------------------|-----------|
| <input type="radio"/> Nat-one | <a href="#">nat-0d3e4203663594174</a> | Public | <input checked="" type="radio"/> Deleted | - | <a href="#">3.212.73.80</a> | 10.0.0.47 |
|-------------------------------|---------------------------------------|--------|------------------------------------------|---|-----------------------------|-----------|

\*Nat Gateway Deleted

## Test S3 Access

```
~~~~~
~..-./
  /m/'
Last login: Tue Sep 23 12:02:34 2025 from 10.0.0.114
[ec2-user@ip-10-0-1-176 ~]$ aws s3 ls
2025-09-23 11:50:14 bucket-one-74148325
2025-09-23 11:50:35 bucket-two-4568563
[ec2-user@ip-10-0-1-176 ~]$ aws s3 ls
2025-09-23 11:50:14 bucket-one-74148325
2025-09-23 11:50:35 bucket-two-4568563
[ec2-user@ip-10-0-1-176 ~]$ aws s3 ls
2025-09-23 11:50:14 bucket-one-74148325
2025-09-23 11:50:35 bucket-two-4568563
[ec2-user@ip-10-0-1-176 ~]$
```

Now I am able to access s3 without internet